

- 1 (a) A student wants to determine the volume of a cylindrical rod by measuring its height and diameter.

- (i) The rod has the following dimensions.

Height = 12.6 ± 0.2 cm

Diameter = 42.61 ± 0.02 mm

Determine the volume of the rod and its associated uncertainty.

volume = $\pm$ cm^3 [3]

- (ii) The student tried to determine the volume of the rod by fully submerging it into a measuring beaker partially filled with water. The measuring beaker has an uncertainty of 1 cm^3 .

The student claims that this method is more accurate in determining the volume of the rod. Comment.

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..... [1]

(iii) Suggest a way to improve the accuracy of method mentioned in (a)(i).

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 [1]

- (b) A boat is adrift at sea. An island is situated 45° East of South from the boat and is 50.0 km away from it. The location of the boat is as shown in Fig 1.1. There is a constant ocean drift of 5.0 km h^{-1} in the easterly direction. The boat can achieve a maximum speed of 30.0 km h^{-1} .

Determine, using scale drawing **or otherwise**, the direction and speed that the boat has to travel such that it will reach the island in exactly 2 hours.

Boat



Fig 1.1

direction = $^\circ$

speed = km h^{-1} [3]

[Turn over